DISTINGUISHING HOMEOMORPHIC 4-MANIFOLDS BY SLICING AND THE EXOTIC $S^2 \times S^2$ PROBLEM

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Abstract. Lidman and Piccirillo constructed the first pair of closed 4-manifolds with isomorphic integer cohomology rings distinguished by unconstrained knot slicing: spin rational homology 4-spheres B and W with $H_1 = \mathbb{Z}/2$ in which the figure-eight knot is slice and not slice, respectively. We show that the natural upgrade of their theorem — a *homeomorphic* such pair — is equivalent to the exotic $S^2 \times S^2$ problem: for every admissible Luttinger-surgery variant V' of their key piece (we classify the admissible parameters), $W' = V'/\sigma$ is homeomorphic to B if and only if the symplectic double $Z' = V' \cup_\sigma V'$ is simply connected, in which case Z' is an exotic $S^2 \times S^2$; if moreover the piece V' itself is simply connected, their regluing additionally yields a simply connected exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$. The proof combines the Hambleton–Kreck classification with a rigidity analysis of the surgery parameters. We then localize what stands between the construction and the wall: building a fully explicit model of the relevant surface-bundle group from the trefoil monodromy, we verify by machine that all 72 uncorrected candidate presentations of $\pi_1(V')$ collapse to the trivial group, that the collapse is carried entirely by four monodromy relations broken by the surgery tori, and that across 120 sampled correction words both behaviors — collapse to the trivial group and resistance to every enumeration — are generic. The same tools verify that the group theory of Akhmedov–Park's claimed exotic $S^2 \times S^2$ (arXiv:1005.3346) is correct given the presentation of their Lemma 8, isolating the open content of that sixteen-year-old preprint in the geometric derivation of its meridian and framing words. The computation itself — deriving the honest based-loop data and deciding the groups — is carried out in a companion paper.

1. Introduction

One approach to distinguishing smooth structures, going back to Casson, is to find a knot that is smoothly slice in one 4-manifold and not in another with the same underlying topology. Lidman and Piccirillo [LP25] recently ran the first successful version of this program at the level of cohomology rings:

Theorem 1.1 ([LP25, Theorem 1])**.** *There are spin rational homology four-spheres B and W with $H_1 = \mathbb{Z}/2$ and isomorphic integer cohomology rings such that the figure-eight knot 4_1 is slice in B but not in W.*

Here B is the Kawauchi manifold [Kaw09] and $W = V/\sigma$, where V is a symplectic homology $S^2 \times D^2$ built from a genus-2 surface bundle R over a once-punctured torus by two Luttinger surgeries, and σ is a free involution of $\partial V = S^3_0(Q)$, Q the square knot. The natural strengthening — replace "isomorphic cohomology rings" by "homeomorphic", producing the first homeomorphic-but-not-diffeomorphic pair distinguished by unconstrained slicing — requires exactly $\pi_1(W') = \mathbb{Z}/2$ for a variant W' of the construction. Our first result says this is not an incremental improvement but a wall, and explains, we believe, why [LP25] stops at cohomology rings.

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To state it, call a 4-manifold V' an *admissible variant* of V if it is obtained from R by Luttinger surgeries as in Proposition 1.3 below (these are exactly the surgeries on the tori T_α, T_β and parallel copies producing a homology $S^2 \times D^2$; the LP manifold is the case $(m,n) = (0,0)$ with no extra tori). Set $W' = V'/\sigma$ and $Z' = V' \cup_\sigma V'$.

Theorem 1.2 (Reduction)**.** *Let V' be any admissible variant. Then:*

(a) *$W' \cong_{\text{homeo}} B$ if and only if $\pi_1(Z') = 1$.*
(b) *If $\pi_1(Z') = 1$, then Z' is homeomorphic to $S^2 \times S^2$ and not diffeomorphic to it, and the pair (B, W') is homeomorphic and non-diffeomorphic, distinguished by the sliceness of 4_1.*
(c) *If $\pi_1(V') = 1$ (which implies $\pi_1(Z') = 1$), then in addition the regluing of* [LP25, Theorem 2] *applied to Z' yields a simply connected 4-manifold homeomorphic but not diffeomorphic to $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.*

An exotic $S^2 \times S^2$ — or an exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ — is a long-standing open problem; the only claim in the literature, by Akhmedov and Park [AP10], has remained an unpublished preprint since 2010 and is cited as such by the current literature [SS23, BSS24], while [LP25] re-proves the strictly weaker "nonstandard cohomology $S^2 \times S^2$" statement (their Theorem 8) and describes the known examples as "presumably not simply connected". Theorem 1.2 thus cuts both ways: the slicing upgrade cannot be obtained without solving the exotic $S^2 \times S^2$ problem inside a σ-symmetric double, and conversely any solution of the π_1-killing problem in the LP frame is worth three theorems at once — strictly dominating the Akhmedov–Park frame, whose payoff is the exotic $S^2 \times S^2$ alone, at zero marginal cost, since surgeries in $\operatorname{int} V$ descend to the quotient automatically.

The proof of Theorem 1.2(a) rests on the topological classification of 4-manifolds with finite cyclic fundamental group: by Freedman and Hambleton–Kreck [Fre82, HK88, HK93] (see [Ham08, Theorem 5.1] for the statement in this form, and [BSS24] for its $b_2 = 0$ case), a closed oriented topological 4-manifold with $\pi_1 = \mathbb{Z}_n$ is determined up to homeomorphism by its intersection form, w_2-type, and Kirby–Siebenmann invariant; in particular the smooth spin rational homology 4-sphere with $\pi_1 = \mathbb{Z}/2$ is unique up to homeomorphism. The input on the smooth side is a rigidity statement for the surgery parameters:

Proposition 1.3 (Rigidity of the surgery parameters)**.** *Any Luttinger surgeries on $T_\alpha, T_\beta \subset R$ producing a homology $S^2 \times D^2$ have coefficient ± 1, and the surgery directions equal α', β' modulo fiber classes. Conversely, for every $m, n \in \mathbb{Z}$ the directions $\beta' e^m, \alpha' c^n$ with coefficients ± 1 (and, optionally, ± 1-surgeries along parallel Lagrangian copies of T_α, T_β with fiber directions) yield $H_1(V') = 0$ and preserve: the spin condition, symplecticity, $\chi = 2$, $H_2(V') = \mathbb{Z}\langle F \rangle$, the descent of σ, and the hypotheses of* [SS23, Theorem 1.4] *for the double. In particular* [LP25, Lemma 7] *holds verbatim for W', and the slicing obstruction for 4_1 in W' persists.*

The second half of the paper concerns what stands between the construction and $\pi_1(V') = 1$, and makes the obstruction quantitative. In Section 2 we build a fully explicit finite presentation of $\pi_1(R)$ with no genus-2 mapping class computations: since Q is the square knot, the closed genus-2 fiber splits along the connect-sum circle and the entire bundle structure is generated by the trefoil monodromy

$$h \colon x \mapsto y^{-1}, \qquad y \mapsto yx$$

on $F_2 = \langle x, y \rangle$, together with the handle swap. All structural identities (that h is the composite of the two standard twists; that it fixes the boundary word $[x, y]$ exactly; that h^6

is the boundary Dehn twist and h^3 the hyperelliptic involution; that Lidman–Piccirillo's commutator factorization $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$ is the literal automorphism identity $\tilde{\psi}\tilde{\varphi}\tilde{\psi}^{-1}\tilde{\varphi}^{-1} = h * h^{-1}$) are certified by machine; the scripts are ancillary files.

With the model in hand, Section 5 reports three computations (GAP 4.16, coset enumeration; all runs reproduce from the ancillary scripts in seconds):

(E1) All 72 *uncorrected* candidate presentations of $\pi_1(V')$ — over all sign and ordering conventions, the LP directions, the mixed directions $m, n \in \{-1, 1\}$, and the parallel-tori scheme — collapse to the trivial group, while the controls ($\pi_1(R)$ alone: $H_1 = \mathbb{Z}^2$; either surgery alone: $H_1 = \mathbb{Z}$) do not. If these presentations were honest, LP's own Z would already be an exotic $S^2 \times S^2$; they are therefore over-determined, and the failure is localized in exactly four monodromy relations, those carried by transport annuli that the surgery tori must intersect.

(E2) Dropping the four broken relations outright leaves $H_1 = \mathbb{Z}^2$: the corrections are indispensable already for homology, so no certificate can bypass them. Across 120 sampled correction words drawn from the geometrically available insertions, 58 give the trivial group and 55 resist enumeration — not visibly trivial — while 7 are homologically infeasible: *both behaviors are generic.* The value of $\pi_1(V')$ — and with it, by Theorem 1.2, the existence of an exotic $S^2 \times S^2$ — is carried entirely by the based-loop correction words.

(E3) From the presentation printed in [AP10, Lemma 8], coset enumeration confirms $\pi_1(M_n^p) = \mathbb{Z}/p$ for $(n, p) \in \{(1, 1), (2, 1), (3, 1), (1, 2), (1, 3)\}$: the group theory of [AP10, Theorem 9] is correct *given* Lemma 8. What remains open in [AP10] is precisely the geometric derivation of the six meridian/framing words of their Lemma 7 — the same kind of based-loop data as in (E2), and not machine-checkable from a figure.

We read (E2) as the formal content of the field's sixteen-year hesitation over [AP10]: at this level of explicitness, nothing short of an honest based diagram decides the group, and both outcomes are unexceptional. The flip side is constructive: the model reduces the wall to a finite, well-posed computation — derive the correction words for the broken monodromy relations (four over the embedded curves; a minimal based diagram realizes three) and the two based surgery relations from an octagon picture of the fiber — after which the ancillary harness decides $\pi_1(V')$, and with it three open problems, in seconds.

Finally, the geometric dual of T_α requires a small correction to the proof of [LP25, Lemma 5] (Remark 3.1), which is used throughout; and the based-loop computation posed by (E2) — deriving the honest meridian, lasso and Lagrangian-framing words and deciding the groups — is carried out in the companion paper [W26b].

2. The explicit model of $\pi_1(R)$

We recall the construction of [LP25, Section 1]. The 0-surgery $S^3_0(Q)$ on the square knot $Q = 3_1\#\overline{3_1}$ fibers over S^1 with closed genus-2 fiber F and monodromy conjugate to $abd^{-1}e^{-1}$, a positive Dehn twist along each of the chain curves a, b, c, d, e of [LP25, Figure 1] being denoted by the same letter. Let φ be the involution of F exchanging $a \leftrightarrow e$, $b \leftrightarrow d$ and preserving c setwise. Since $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$ in the mapping class group, the F-bundle R over the once-punctured torus with monodromy ab along β and φ along α has $\partial R = S^3_0(Q)$.

Convention 2.1. $[u, v] = uvu^{-1}v^{-1}$. Automorphisms act on the left and compose right-to-left: fg means "g first". Mapping classes act on π_1 only up to inner automorphism; every presentation below depends on a choice of automorphism lifts, see Remark 2.6.

2.1. **The trefoil monodromy.** Let $\Sigma_{1,1}$ be a once-punctured torus with basepoint on the boundary, $\pi_1(\Sigma_{1,1}) = F_2 = \langle x, y\rangle$, boundary word $[x, y]$, and let the core curves realizing x, y be $a \simeq x$ and $b \simeq y$, so $a \cdot b = 1$. For a suitable orientation convention the twists act by

$$T_a\colon (x, y) \mapsto (x,\ yx), \qquad T_b\colon (x, y) \mapsto (xy^{-1},\ y).$$

Define $h := T_a \circ T_b$ (T_b first).

Lemma 2.2. *$h(x) = y^{-1}$, $h(y) = yx$, and:*

1. $h([x, y]) = [x, y]$ exactly *(not merely up to conjugacy);*
2. $h^6 = \mathrm{conj}_{[x,y]^{-1}}$*, the full boundary Dehn twist;*
3. h^3 *acts on* H_1 *as* $-\mathrm{id}$*;*
4. *on* $H_1(\Sigma_{1,1}) = \mathbb{Z}^2$*,* h *acts by* $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 1 \end{smallmatrix}\right)$*: determinant* 1*, trace* 1*, order* 6*.*

Consequently h is the fibered monodromy of a trefoil: it is a product of single positive twists along dual non-separating curves, fixes the boundary pointwise (reflected in (1)), and is freely periodic of period 6 with h^6 the boundary twist (2), with the hyperelliptic involution at h^3 (3).

Proof. Finite computations in F_2, certified by the ancillary scripts `monodromy_check2.g` and `model_check3.g`, and easily reproduced by hand. For (1):

$$h([x, y]) = [y^{-1}, yx] = y^{-1} \cdot yx \cdot y \cdot x^{-1}y^{-1} = xyx^{-1}y^{-1} = [x, y].$$

□

Remark 2.3 (Chirality)**.** Whether h or h^{-1} ($h^{-1}\colon x \mapsto xy,\ y \mapsto x^{-1}$) is the right-handed trefoil is a convention we will not need: Q is the connected sum of a trefoil and its mirror, and exchanging the chirality assignment amounts to the handle relabeling $(x, y) \leftrightarrow (r, s)$ below, under which every construction and every computation in Section 5 is symmetric.

2.2. **The closed fiber and the bundle group.** The fiber of $Q = 3_1\#\overline{3_1}$ is the boundary connected sum of the two trefoil fibers, and the closed fiber F of $S^3_0(Q)$ splits along the connect-sum circle into two once-punctured tori. Writing $\pi_1(F) = \langle x, y, r, s \mid [x, y][r, s]\rangle$ with (x, y) carried by the left handle and (r, s) by the right, the dictionary with [LP25, Figure 1] is

$$a \sim x, \quad b \sim y, \quad e \sim r, \quad d \sim s, \qquad c \sim xr, \quad z \sim ys,$$

where the last two identify the φ-invariant homology classes $[c] = [a] + [e]$ and $[z] = [b] + [d]$; the specific words xr and ys are representative choices (see Remark 2.6 and Section 5). Define

$$\tilde\varphi\colon\ x \leftrightarrow r,\ y \leftrightarrow s \qquad \text{and} \qquad \tilde\psi := h * \mathrm{id}\colon\ (x, y, r, s) \mapsto (y^{-1},\ yx,\ r,\ s),$$

lifting φ and ab respectively; $\tilde\psi$ is well defined on the closed-fiber group because h fixes the boundary word exactly (Lemma 2.2(1)).

Lemma 2.4. (1) *$\tilde\psi$ fixes the relator $[x, y][r, s]$ exactly; $\tilde\varphi$ sends it to its $[x, y]$-conjugate. In particular both define automorphisms of $\pi_1(F)$.*

(2) $\tilde\varphi^2 = \mathrm{id}$*, and* $\tilde\psi\,\tilde\varphi\,\tilde\psi^{-1}\,\tilde\varphi^{-1} = h * h^{-1}\colon (x, y, r, s) \mapsto (y^{-1},\ yx,\ rs,\ r^{-1})$*.*

Proof. Machine-certified (`model_check3.g`); (2) also follows from block algebra: $\tilde{\varphi}(h * \mathrm{id})^{-1}\tilde{\varphi} = \mathrm{id} * h^{-1}$, so the commutator is $(h * \mathrm{id})(\mathrm{id} * h^{-1}) = h * h^{-1}$. ☐

Lemma 2.4(2) realizes the mapping-class factorization $abd^{-1}e^{-1} = (ab)\varphi(ab)^{-1}\varphi^{-1}$ of [LP25] as a literal identity of automorphisms: $h * h^{-1}$ is the 0-surgery monodromy of the square knot in this model (the left handle twisted by the trefoil, the right by its mirror).

Proposition 2.5 (The presentation)**.**

$$\pi_1(R) = \big\langle x, y, r, s, \alpha, \beta \;\big|\; [x,y][r,s], \\ \alpha g \alpha^{-1} = \tilde{\varphi}(g),\ \beta g \beta^{-1} = \tilde{\psi}(g) \quad (g \in \{x,y,r,s\})\big\rangle.$$

Proof. R fibers over the once-punctured torus T_0 with aspherical fiber F; the long exact sequence of the fibration reduces to $1 \to \pi_1(F) \to \pi_1(R) \to \pi_1(T_0) \to 1$, and $\pi_1(T_0) = F_2\langle \alpha, \beta\rangle$ is free, so the sequence splits and $\pi_1(R) = \pi_1(F) \rtimes F_2$ with respect to the chosen automorphism lifts $\tilde{\varphi}, \tilde{\psi}$ of the two monodromies. The displayed presentation is the standard presentation of such a semidirect product. ☐

Remark 2.6 (Lift freedom; where the based-loop indeterminacy lives)**.** A mapping class determines its action on $\pi_1(F)$ only up to inner automorphisms. Replacing $\tilde{\varphi}, \tilde{\psi}$ by other lifts changes the presentation of Proposition 2.5 by the substitution $\alpha \mapsto \alpha w$, $\beta \mapsto \beta w'$ $(w, w' \in \pi_1(F))$ — an isomorphism of $\pi_1(R)$, but one that changes the *words* that geometric objects (meridians, surgery directions) are represented by. All presentations of the surgered manifolds in Section 5 are therefore *candidates*, well defined only once based representatives are fixed; quantifying the sensitivity of $\pi_1(V')$ to that choice is precisely the point of experiment (E2).

Remark 2.7 (Convention robustness)**.** Replacing the monodromy lift $\tilde{\psi}$ by $\tilde{\psi}^{-1}$ (the holonomy-direction convention) is the substitution $\beta \mapsto \beta^{-1}$, which converts the surgery relation $[z,\alpha]^{\varepsilon}\beta = 1$ into $[z,\alpha]^{-\varepsilon}\beta = 1$: an ε-flip, and the computational grid of Section 5 sweeps all ε. Together with Remark 2.3, the collapse of the LP cases below is independent of every orientation and composition convention made here; the mixed-direction and parallel-tori cases (whose convention variants differ additionally by word order and conjugation) are run in the conventions stated.

3. Rigidity of the surgery parameters

Recall from [LP25, Section 1] the two Lagrangian tori: T_β, the sub-bundle with fiber e over β, and T_α, the sub-bundle with fiber c over α (the monodromy ab fixes e pointwise; φ preserves c with orientation). Their meridians are commutators in the complement $C := R \setminus (T_\alpha \cup T_\beta)$: $\mu_{T_\beta} \simeq [z, \alpha'']$ and $\mu_{T_\alpha} \simeq [d, \beta'']$ (see Remark 3.1), where α'', β'' project to α, β. More generally, for the parallel-copy families below, every relevant meridian is such a commutator: the geometric-dual construction applies verbatim to each parallel copy, whose dual sub-bundle torus meets it once and exhibits the meridian as the corresponding commutator.

Remark 3.1 (The geometric dual of T_α)**.** The proof of [LP25, Lemma 5] prints the dual as (b, β''); this cannot close up into a torus, since ab does not preserve b even homologically (the only twist in ab affecting b is T_a, and $a \cdot b = 1$, so $[ab(b)]$ has $[a]$-coefficient ± 1 in every convention). The curve dual to c that ab *does* fix — pointwise, being disjoint from $a \cup b$, exactly as [LP25, p. 3] observe for e — is d; the dual is therefore $T^*_\alpha = (d, \beta'')$, with μ_{T_α}

freely homotopic to $[d, \beta'']$. Since φ exchanges $b \leftrightarrow d$, the slip is a natural one, and no result of [LP25] is affected: their arguments use only that each meridian is a commutator of a fiber curve with a curve projecting to the base. At the π_1 level, however, the correction matters, and it is used throughout this paper and the companion [W26b].

Lemma 3.2. *Let $C = R\backslash\nu(\mathcal{T})$ for any finite disjoint union $\mathcal{T}$ of tori from the two admissible sub-bundle families. Then the inclusion induces an isomorphism*

$$H_1(C) \xrightarrow{\cong} H_1(R) = \mathbb{Z}^2\langle\alpha', \beta'\rangle.$$

In particular every fiber class vanishes in $H_1(C)$.

Proof. First, $H_1(R) \cong \mathbb{Z}^2$: by Proposition 2.5, $H_1(R)$ is the direct sum of $\mathbb{Z}^2\langle\alpha, \beta\rangle$ and the coinvariants of $H_1(F) = \mathbb{Z}^4$ under the subgroup generated by $\tilde{\varphi}_*$ and $\tilde{\psi}_*$; the stacked matrix $(\Phi - I; \Psi - I)$ has Smith normal form $\mathrm{diag}(1, 1, 1, 1)$ (ancillary script `model_check3.g`), so the coinvariants vanish. (Concretely: $\Psi - I$ is invertible on the (x, y)-block, killing x, y; then the swap relations kill r, s.)

For the complement, excision and the Künneth formula for $(\nu(T), \partial\nu(T)) \cong T^2 \times (D^2, \partial D^2)$ give $H_1(R, C) = 0$ and $H_2(R, C) \cong \bigoplus_{T\in\mathcal{T}} \mathbb{Z}$, generated by the normal disks, whose boundaries are the meridians. The homology sequence of the pair,

$$H_2(R, C) \xrightarrow{\partial} H_1(C) \to H_1(R) \to H_1(R, C) = 0,$$

therefore presents $H_1(R)$ as the quotient of $H_1(C)$ by the images of the meridians; each meridian is a commutator in $\pi_1(C)$, hence trivial in $H_1(C)$, so the map $H_1(C) \to H_1(R)$ is an isomorphism. □

Proof of Proposition 1.3. Forcing. A Luttinger ($1/k$-log) surgery on T_β along an embedded direction, i.e. a primitive class $p\beta' + qe$ on T_β ($\gcd(p, q) = 1$), fills the boundary of $\nu(T_\beta)$ so that the class $\mu_{T_\beta} + k(p\beta' + qe)$ dies (orientation conventions are absorbed into the sign of k). By Lemma 3.2, in H_1 this relation reads $kp\,\beta' = 0$, since the meridian is a commutator and e is a fiber class. Together with the corresponding relation $k'p'\,\alpha' = 0$ from T_α,

$$H_1(V') \;=\; \mathbb{Z}^2/\langle kp\,\beta',\, k'p'\,\alpha'\rangle \;=\; \mathbb{Z}/|kp| \oplus \mathbb{Z}/|k'p'|,$$

which vanishes iff $|kp| = |k'p'| = 1$: the coefficients are ±1 and the directions have base component ±1, i.e. equal β', α' modulo fiber classes. Purely-fiber directions ($p = 0$) impose no relation and leave $H_1 = \mathbb{Z}^2$.

Realization. Conversely, for coefficients ±1 and directions $\beta' e^m, \alpha' c^n$ the same computation gives $H_1(V') = 0$; adding ±1-surgeries along parallel copies with fiber directions adds only trivial H_1-relations and keeps $H_1(V') = 0$. The admissible parallel families are constrained by the intersection pattern of the fiber curves: with $[c] = [a] + [e]$ and $[z] = [b] + [d]$ one has $c \cdot e = 0$, $c \cdot d = z \cdot e = d \cdot e = 1$, $z \cdot d = 0$, so the maximal disjoint families are $\{c\text{-fibered over } \alpha\text{-parallels}\} \cup \{e\text{-fibered over } \beta\text{-parallels}\}$ (the LP family) or the complementary $\{z/\alpha\} \cup \{d/\beta\}$ family, but no mixture.

Preserved structure. Each direction above is a primitive class realized by an embedded curve on the corresponding Lagrangian torus, so the surgeries are Luttinger surgeries and the result carries a symplectic structure [Lut95, ADK03]; χ is unchanged. The argument of [LP25, Lemma 5] now applies verbatim: $H_1(V') = 0$ and $\partial V'$ connected give $H_3(V') = 0$ and $H_2(V')$ free; $\chi(V') = 2$ gives $H_2(V') \cong \mathbb{Z}$, generated by a fiber F disjoint from all surgery tori; $F^2 = 0$; since $H_1(V') = 0$ we get $H_2(V'; \mathbb{Z}/2) \cong \mathbb{Z}/2\langle F\rangle$, and evaluation against it detects $H^2(V'; \mathbb{Z}/2)$, so $\langle w_2, F\rangle = F \cdot F = 0$ forces $w_2(V') = 0$: V' is spin. All surgeries take place in $\operatorname{int} V$, so $\partial V' = \partial V = S^3_0(Q)$ and the free involution σ descends; the proof of [LP25, Lemma 7] applies verbatim and $W' = V'/\sigma$ is a spin rational homology 4-sphere with $H_1(W') = \mathbb{Z}/2$.

Persistence of the obstruction input. The double $Z' = V' \cup_\sigma V'$ is obtained from the genus-2 bundle $R \cup_\sigma R$ over a genus-2 surface by Luttinger surgeries on disjoint Lagrangian tori; the fiber F and the section $\hat{\Gamma}$ coming from a fixed point of φ ([LP25, Lemma 6]) form a hyperbolic pair, and all surgery tori miss a fiber (they lie over curves in the base, which miss a point) and miss the two fixed-point sections (the fiber curves c, e, z, d and their parallel copies can be isotoped off the two fixed points of φ, exactly as in [LP25, Lemma 6]). These are the hypotheses of [SS23, Theorem 1.4] as used in [LP25, proof of Theorem 1, p. 6], which constrains neither the surgery coefficients nor the directions. Hence the slicing obstruction for 4_1 in W' persists; the details are completed in the proof of Theorem 1.2(b) below. □

4. The reduction theorem

Proof of Theorem 1.2. (a). The double cover $Z' \to W'$ gives a short exact sequence $1 \to \pi_1(Z') \to \pi_1(W') \to \mathbb{Z}/2 \to 1$, with $\pi_1(Z') \hookrightarrow \pi_1(W')$ *injective* by covering theory.

($\Leftarrow$) If $\pi_1(Z') = 1$ then $\pi_1(W') \cong \mathbb{Z}/2$. Now W' and B are closed, oriented, *smooth* spin 4-manifolds with $\pi_1 = \mathbb{Z}/2$ (Proposition 1.3 for W'; [Kaw09, LP25] for B) and both are rational homology spheres, so both intersection forms on H_2/tors are trivial, both w_2-types are type (II), and both Kirby–Siebenmann invariants vanish (smoothness; alternatively $\mathrm{KS} \equiv \sigma/8 \pmod 2$ for spin topological 4-manifolds). By the classification of closed oriented topological 4-manifolds with finite cyclic fundamental group — [HK93, Theorem C]: π_1, the intersection form on H_2/Tors, the w_2-type and KS are a complete set of invariants (extending [HK88, Theorem B] from odd order; see also [Ham08, Theorem 5.1] and [Tei92]) — we get $W' \cong_{\text{homeo}} B$. In fact, by the $b_2 = 0$ case as stated in [BSS24] there are exactly two such topological manifolds for each cyclic group, one spin and one non-spin; the spin one is the spun lens space, so $W' \cong_{\text{homeo}} B \cong_{\text{homeo}} L_2$.

($\Rightarrow$) A homeomorphism gives $\pi_1(W') \cong \pi_1(B) = \mathbb{Z}/2$, and the injection above forces $|\pi_1(Z')| = 1$.

(b). Assume $\pi_1(Z') = 1$.

Step 1: Z' is homeomorphic to $S^2 \times S^2$. Z' is closed, simply connected, spin (it double covers the spin manifold W'; alternatively glue the spin structures of the two copies of V'), with $\chi(Z') = 2\chi(V') - \chi(S^3_0(Q)) = 4$, so $b_2(Z') = 2$. The classes of the fiber F and of the closed section $\hat{\Gamma} = \Gamma \cup_\sigma \Gamma$ satisfy $F^2 = 0$ and $F \cdot \hat{\Gamma} = 1$; since the form is even (spin), a change of basis $\hat{\Gamma} \mapsto \hat{\Gamma} - \frac{1}{2}(\hat{\Gamma}^2)F$ exhibits the intersection form as the hyperbolic form H. By Freedman's classification [Fre82] (even forms are realized by a unique topological manifold), $Z' \cong_{\text{homeo}} S^2 \times S^2$.

Step 2: Z' is not diffeomorphic to $S^2 \times S^2$. This is the argument of [LP25, Theorem 8], which is π_1-agnostic: $R \cup_\sigma R$ is a non-trivial genus-2 surface bundle over a genus-2 surface, hence aspherical, hence minimal and neither rational nor ruled, so its Kodaira dimension satisfies $\kappa \neq -\infty$; Luttinger surgery preserves κ [HL12], and Z' is minimal (it is spin), so $\kappa(Z') \neq -\infty$. Since κ is a diffeomorphism invariant [Li06] and $\kappa(S^2 \times S^2) = -\infty$, Z' is not diffeomorphic to $S^2 \times S^2$.

Step 3: the pair (B, W'). By (a), $W' \cong_{\text{homeo}} B$. The knot 4_1 is slice in B by construction [Kaw09, LP25]. It is not slice in W': following [LP25, Section 2], since W' is spin the relative Rokhlin theorem [Klu21, Theorem 2] and $\mathrm{Arf}(4_1) = 1$ rule out a null-homologous slice disk; a homologically essential slice disk makes the 0-trace $X_0(4_1)$ embed in W' with

H_2-essential image, producing a square-zero essential torus $T \subset W'$ (cap a Seifert surface with the disk); since $X_0(4_1)$ is simply connected, T lifts to an essential square-zero torus in Z'; and Proposition 1.3 verifies for Z' the hypotheses of [SS23, Theorem 1.4], which forbids such a torus. Sliceness of 4_1 distinguishes the smooth structures, so W' is homeomorphic and not diffeomorphic to B.

(c). Assume $\pi_1(V') = 1$. The double Z' and the reglued manifold $Z'' = V' \cup_{f\circ\sigma} V'$ (with f the boundary self-homeomorphism of [LP25, Figure 3]) are unions of two copies of V' along $S^3_0(Q)$; by van Kampen each fundamental group is a quotient of $\pi_1(V') * \pi_1(V') = 1$, so $\pi_1(Z') = \pi_1(Z'') = 1$ and parts (a)–(b) apply. (This is where the hypothesis $\pi_1(V') = 1$ is genuinely used: the regluing twists the van Kampen data by f_*, and no implication from $\pi_1(Z') = 1$ alone is claimed; see Remark 4.1.) The framing-parity change makes Z'' non-spin with $b_2 = 2$ and signature 0, so its intersection form is $\langle 1\rangle \oplus \langle -1\rangle$ and Freedman gives $Z'' \cong_{\text{homeo}} \mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. The relative invariants $\Psi_{V',\mathfrak{t}_\pm}$ are non-zero exactly as in [LP25, Lemma 9]: Z' is symplectic and its canonical class evaluates ± 2 on a fiber disjoint from the surgery tori — Luttinger surgery does not change this pairing [ADK03, HL12] — so [OS04] applies as there (cf. [Thu76]). By [LP25, Lemma 10] the mixed invariant of Z'' for the line span$\{F\}$ is non-zero; since $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$ (indeed $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2\#D$ for any homology sphere D) admits, for every choice of line, a square-zero splitting along $S^2 \times S^1$ forcing all mixed invariants to vanish ([LP25, proof of Theorem 2], using $HF_{red}(S^2 \times S^1) = 0$), Z'' is not diffeomorphic to $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$. □

Remark 4.1. The implication $\pi_1(V') = 1 \Rightarrow \pi_1(Z') = 1$ is immediate from van Kampen; we do not know whether the converse holds in this setting, since in the amalgam $\pi_1(Z') = \pi_1(V') *_{\pi_1 S^3_0(Q)} \pi_1(V')$ the edge maps need not be injective. The distinction affects only part (c) — parts (a) and (b) use $\pi_1(Z')$ alone, while the regluing twists the van Kampen data by f_* and is controlled here through $\pi_1(V')$ — and it is immaterial in practice: any certification of simple connectivity in this frame proceeds by computing $\pi_1(V')$, which is what the harness of Section 5 does.

Corollary 4.2. *The homeomorphic upgrade of* [LP25, Theorem 1] *within the LP frame is* gated*: it holds for some admissible variant if and only if some admissible double Z' is an exotic $S^2 \times S^2$. Conversely, exhibiting $\pi_1(V') = 1$ for any admissible variant yields simultaneously an exotic $S^2 \times S^2$, a homeomorphic-but-not-diffeomorphic pair distinguished by unconstrained slicing, and a simply connected exotic $\mathbb{CP}^2\#\overline{\mathbb{CP}}^2$.*

5. Computational experiments

All computations are coset enumerations and abelianizations in GAP 4.16 [GAP]; the scripts are provided as ancillary files,[1] and every run below reproduces in seconds on a laptop (the enumeration cap is 4×10^5 cosets; a run that exceeds it is reported as *blowup*: not visibly trivial, never "nontrivial". In that event the (E1) harness would attempt a nontriviality certificate by exhibiting a perfect quotient; no (E1) run needed it, and the blowups of (E2) carry no such certificate).

[1] `pi1_grid.g`, `pi1_v2b.g`, `ap_check.g`, `monodromy_check2.g`, `model_check3.g`; also available at https://github.com/bwuebben/exotic-s2xs2.

5.1. **Candidate presentations.** By Proposition 2.5 and Remark 2.6, a presentation of $\pi_1(V')$ requires, beyond the exact bundle relations, based representatives for the two surgery relations. The *uncorrected candidates* take the naive words suggested by the free homotopy types: filling relations

$$[z,\alpha]^{\varepsilon_1} \cdot \beta\, r^m = 1, \qquad [w,\beta]^{\varepsilon_2} \cdot \alpha\,(xr)^n = 1,$$

with $z \in \{ys, sy\}$, $\varepsilon_i \in \{\pm 1\}$, mixed-direction exponents m, n as in Proposition 1.3, and, in the parallel-tori scheme, the additional fiber-direction relations $[z,\alpha]^{\varepsilon_3} r = 1$, $[w,\beta]^{\varepsilon_4}(xr) = 1$. The grid of (E1) takes $w = y$, the dual printed in [LP25]; the corrected dual $w = s$ of Remark 3.1 is run separately.

5.2. **(E1) The uncorrected candidates collapse universally.** Over all 8 conventions for the LP surgeries, all 32 mixed-direction cases ($m, n \in \{-1, 1\}$), and all 32 parallel-tori cases, *every one of the* 72 *candidate groups coset-enumerates to the trivial group* (script `pi1_grid.g`; total runtime under a second). The four sign cases built instead on the corrected dual $w = s$ (script `pi1_v2b.g`) are likewise all trivial. The controls behave correctly (script `pi1_v2b.g`): the bundle group alone has $H_1 = \mathbb{Z}^2$, and either surgery alone leaves $H_1 = \mathbb{Z}$, so the collapse requires both surgeries even homologically and is not an artifact of the machinery.

If the uncorrected candidates were honest presentations, then already the original LP manifold would satisfy $\pi_1(V) = 1$ and, by Theorem 1.2, Z would be an exotic $S^2 \times S^2$ — which [LP25] explicitly does not claim (their Theorem 8 proves the strictly weaker cohomology statement, and they describe such examples as “presumably not simply connected”). The candidates are therefore over-determined, and the failure is localizable a priori. A monodromy relation $\beta g \beta^{-1} = \tilde{\psi}(g)$ is carried by a transport annulus $g \times \beta$; in the surgered manifold the relation holds only if the annulus avoids the surgery tori. Intersection numbers in R decide this: an annulus over one base curve meets a sub-bundle torus over the transverse base curve above their intersection point, and meets a torus over the *same* base curve along it; in both cases the count is the intersection number of the fiber curves. Using $[c] = [a] + [e]$, $[z] = [b] + [d]$:

relation carried over α or β	obstruction
$x\ (\sim a),\ r\ (\sim e)$ over α, β	$a \cdot c = a \cdot e = e \cdot c = 0:$ clean
$y\ (\sim b)$ over β	$b \cdot c = 1:$ crosses T_α
y over α	$b \cdot c = 1:$ crosses T_α
$s\ (\sim d)$ over α	$d \cdot c = 1:$ crosses T_α
s over β	$d \cdot e = 1,\ d \cdot c = 1:$ crosses T_β, T_α

Exactly the four relations for $y \sim b$ and $s \sim d$ are broken; in V' they hold only with insertions of conjugates of words carried by the glued-in tori (c-, e-, α'-, β'-words). The four relations for x and r are clean. (The count is that of membranes over the *embedded* curves. Like the correction words themselves, it is diagram data: in the companion's diagram, whose based base loops meet the embedded curves minimally, only three relations acquire corrections, each at a single crossing.)

5.3. **(E2) The corrections decide everything.** Two further computations (script `pi1_v2b.g`).

No certificate bypasses the corrections. Dropping the four broken relations outright leaves a group with $H_1 = \mathbb{Z}^2$: the broken relations are indispensable already for $H_1(V') = 0$. In particular no surjection from a clean-relations group onto the honest $\pi_1(V')$ can certify anything — the clean group is not even a homology-correct approximation.

Sensitivity. We sample the four correction words in the form $g_i u_i g_i^{-1}$, with u_i drawn from the torus-carried words assigned by the crossing table above ($\{(xr)^{\pm 1}, \alpha^{\pm 1}\}$ for the relations crossing T_α; $\{r^{\pm 1}, \beta^{\pm 1}\}$ for the s-over-β relation, sampled from its T_β-carried words) and g_i from nine short conjugators, with random signs ε_i; 120 pseudorandom trials with fixed seed 20260709. Outcomes:

$$\textbf{58 trivial}, \qquad \textbf{55 blowup}, \qquad \textbf{7 infeasible}\ (H_1 \neq 0).$$

("Blowup" means the enumeration exceeds the cap: not visibly trivial. The honest corrections preserve $H_1 = 0$, so the 7 infeasible samples only mark the sampler's support as slightly larger than the geometric one.) Both outcomes are generic: the value of $\pi_1(V')$ — and with it, by Theorem 1.2, the existence of an exotic $S^2 \times S^2$ — is carried entirely by the based-loop correction words, with no robustness in either direction. Nothing short of an honest based diagram decides the group.

5.4. **(E3) Verification of the Akhmedov–Park group theory.** The manifold of [AP10] is built from the same $\mathbb{Z}/2$-germ: their model surface $(\Sigma_2 \times \Sigma_3)/\mathbb{Z}_2$ fibers over Σ_2 with monodromy the same two-fixed-point involution class as φ. Their Theorem 9 asserts $\pi_1(M_n^p) = \mathbb{Z}/p$ from the presentation printed in their Lemma 8. Coset enumeration from that presentation (script `ap_check.g`) confirms

$$\pi_1(M_n^p) = \mathbb{Z}/p \quad \text{for} \quad (n,p) \in \{(1,1),(2,1),(3,1),(1,2),(1,3)\}.$$

Thus the group theory of [AP10, Theorem 9] is correct *given* their Lemma 8, and the entire content of the sixteen-year-old open status of [AP10] is the geometric derivation of the six meridian/Lagrangian-framing triples of their Lemma 7 (their fifth and sixth, involving a path that is not a sub-bundle coordinate, being the visibly delicate ones). One cannot machine-check a figure; the referee's task for [AP10] and the based-loop computation posed by (E2) are the same task.

6. Discussion

6.1. **The companion computation.** The model of Section 2 reduces the wall to a finite, fully specified computation: derive, from an explicit based picture of F carrying the curves $a, \ldots, e, z$ and the basepoint, the honest correction words for the broken monodromy relations and the based surgery data, in the style of [AP10, Lemma 7] or Baldridge–Kirk [BK08]. Because the crossing pattern of Section 5 is fixed by the intersection combinatorics, a single based diagram parameterizes the entire admissible family of Proposition 1.3 at once. This computation is carried out in the companion paper [W26b], from a rotation-symmetric octagon model in which every monodromy lift is honestly based and every meridian, lasso and Lagrangian-framing word is derived mechanically from finite combinatorial certificates; the resulting presentations are proved complete and decided by coset enumeration. The outcome is that $\pi_1(V') = 1$ for *all nine* cells $m, n \in \{-1, 0, 1\}$ of the mixed-direction family computed there — including the Lidman–Piccirillo surgery itself — so that, by Theorem 1.2, the double Z of [LP25] is an exotic $S^2 \times S^2$, the pair (B, W) is homeomorphic and non-diffeomorphic, distinguished by the sliceness of 4_1, and the regluing yields a simply connected exotic $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$ — and likewise for every admissible variant. Seven cells are decided by coset enumeration. The remaining two, whose enumerations blow past 10^8 cosets — and which a 56-CPU-hour search had shown to have no nontrivial finite quotient of order at most 10^5, as, it turns out, befits the trivial group — are decided by Knuth–Bendix completion, whose confluent rewriting systems reduce every generator to the identity: enumeration blowup proves nothing in either direction, exactly as (E2) warned. We refer to [W26b] for the verification apparatus supporting that computation.

6.2. **A wall-free rung.** Theorem 1.2 does not gate every strengthening of [LP25]. Their Theorem 3 produces a 4-manifold A homeomorphic to $\mathbb{CP}^2\#5\overline{\mathbb{CP}}^2$ in which the non-vanishing of the mixed invariant for the line span$\{F\}$ survives; a constrained non-slicing statement on A — no genus-one knot with non-trivial Arf invariant bounds a disk in a class d with $d \cdot F \neq 0$ and $d^2 \geq 0$ — appears provable with the tools already in [LP25, SS23, MMP24], while the unconstrained statement is plausibly false (the classes invisible to a line-based invariant, $d \cdot F = 0$ or $d^2 < 0$, are exactly where Norman disks live in the standard manifold). We leave this to future work.

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